Capstone Project Proposal: VeriCBAM

A Data-Driven Decision-Support Tool for Screening Carbon Border (CBAM) Emissions Reports

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1. Background & Problem Statement

Under the European Union's Carbon Border Adjustment Mechanism (CBAM), companies importing carbon-intensive goods such as Cement and Iron & Steel into the EU must report the greenhouse gas (CO_2) emissions generated during production and pay a corresponding carbon fee.

- The Real-World Problem: To prevent companies from under-reporting their emissions to save money, reports must be verified by independent auditors. However, sending human inspectors to physically audit thousands of overseas factories is slow, expensive, and logistically difficult.
- The Data Science Opportunity: Before sending an inspector on an expensive site visit, regulators and auditors need a pre-screening tool that automatically cross-checks a factory's self-reported CO_2 number against independent public data to flag suspicious or unrealistic declarations.

2. Research Question & Project Objectives

Core Research Question: How can we combine basic industrial physics, historical emissions records, air pollutant ratios, and satellite observations into an explainable Data Science model that reliably flags inconsistent CO_2 declarations while minimizing false alarms?

- Objective 1 (Multi-Source Data Integration): Collect and align real-world factory emissions, production capacities, and satellite air-quality data for 30 major cement and steel plants.
- Objective 2 (Multi-Stream Anomaly Detection): Build five complementary checks that test a factory's declared emissions from different angles (physical minimums, technology type, historical trends, co-pollutants, and satellite plumes).
- Objective 3 (Explainable Risk Scoring): Combine these five checks into an interpretable risk score that tells a human auditor why a report looks suspicious—or explicitly states "Insufficient Evidence" when data is missing or cloudy.

3. Proposed Methodology: How VeriCBAM Works

Instead of acting as a "black box," VeriCBAM evaluates each emissions report using five intuitive evidence checks:

1. Physical Minimum Check: Producing 1 tonne of cement or steel requires a known minimum amount of raw material and fuel. If a factory declares less CO_2 than is physically possible for its size and technology, it is flagged immediately.
2. Production Technology Check: Cross-checks the factory against global industry registries to ensure a high-emission coal blast furnace is not falsely claiming to be a low-emission scrap recycling plant.
3. Historical Trend Check: Compares the factory's newly declared CO_2 against its own past multi-year average to detect sudden, unexplained drops in reported emissions.
4. Co-Pollutant Cross-Check (NO_x vs. CO_2): High-temperature furnaces produce nitrogen oxides (NO_x) alongside CO_2. If a factory claims its CO_2 dropped sharply while its NO_x pollution stayed high, it indicates potential under-reporting.
5. Satellite Activity Check (Copernicus Sentinel-5P): Uses free European Space Agency satellite data to check whether an active nitrogen dioxide (NO_2) plume is visible over the plant during the reporting year.

- Combining the Evidence (Data Science Models):
- Explainable Bayesian Fusion Model: Combines the five checks into a transparent probability score (0% to 100% risk) and weights each source by its data reliability (for example, reducing the weight of satellite data when clouds block the view).
- Supervised Logistic Regression Model: Trained with 5-fold cross-validation (grouped by facility so the model is always tested on unseen factories) to rank declarations from highest to lowest audit priority.

4. Datasets & Evaluation Plan

- Public Datasets Used:

1. European Environment Agency (E-PRTR Registry): 177 real annual emissions records (2018–2023) across 30 European cement and steel facilities (15 Cement, 15 Steel across 10 countries).
2. Global Energy Monitor (GEM): Independent database of factory production capacities and furnace technologies.
3. Copernicus Data Space (Sentinel-5P): Satellite atmospheric NO_2 measurements over each facility.

- Validation Benchmark (325 Test Cases): Because real fraudulent reports are confidential, the models will be tested on a benchmark of 325 scenarios combining the 177 real compliant factory-years with 148 simulated under-reporting cases (ranging from subtle 5%–10% reductions to severe 30%–50% understatements and technology misclassifications). Models will be compared using ROC-AUC, Precision-Recall, False Positive Rate, and Ablation Studies (testing whether combining all five sources beats using any single source alone).

5. Expected Deliverables

1. Modular Python Pipeline (`src/`): Clean, reproducible Python code for data ingestion, feature extraction, and risk modeling.
2. Interactive Streamlit Web Dashboard: A user-friendly visual application where an auditor can select any facility on a map, inspect the five evidence checks side-by-side, see the step-by-step risk breakdown, and export a 1-page Audit Dossier.
3. Final Master Thesis & Evaluation Report: Complete academic analysis comparing single-source vs. multi-source screening performance.